Customer Segmentation with RFM and K-Means, plus Customer Lifetime Value

A summary of my Master of Data Science dissertation, with a post-dissertation extension

Samuel Lim Wei Howe
Master of Data Science, HELP University, 2023

Dissertation title: *An AI-Based Approach of Customer Data Segmentation Based on K-Means Clustering Algorithms*
Industry partner: a Malaysian health-products retailer (name withheld under a non-disclosure agreement)
Companion code: my public Jupyter notebook for this project (Customer Segmentation with K-Means). Section signs (§) in this report point to the matching notebook sections.

About privacy in this report

The data behind this work belongs to the partner and stays private. Everything here is reported so that no customer and no company detail can be identified:

- no customer names, IDs, contact details, invoices or dates of individual purchases;
- the company, its products and brands are not named: products appear as Line 1, Line 2 … and regions as Region 1, Region 2 …;
- money is never shown in RM: sales are given as shares, and customer value as an index or ranking;
- the figures are counts, percentages and statistics taken from my dissertation's published results;
- the Customer Lifetime Value extension was built and tested on synthetic data with the same structure. No real CLTV results are reported.

Contents

1 Summary at a glance
2 Introduction
3 What earlier studies showed
4 How I approached it
5 The data and how I cleaned it
6 RFM features and K-Means modelling
7 Who the customers are
8 A second scaling: Standard + MinMax
9 A rule-based alternative: RFM quantile scores
10 What they buy: product segmentation
11 Where they buy: regional analysis
12 Conclusions of the dissertation
13 Further work: Customer Lifetime Value
14 Business findings and recommendations
15 How the data is protected
A Appendix A: Report ↔ notebook map
B Appendix B: Key numbers fact sheet
C Appendix C: References

1 Summary at a glance

Notebook: whole notebook; key numbers in Appendix B

62,766 transaction lines, 17 fields, 2017 – 2022	**9,227** customers segmented	**435** hidden duplicate rows removed	**k = 6** K-Means segments, plus 9 rule-based	**~45%** of customers lapsed for 1.5+ years

For my dissertation, I worked with a Malaysian retailer that sells health products online and through branches. Its marketing team had years of customer transactions but no way to see who its customers were, which ones were drifting away, or where to grow next. I cleaned the data, described every customer by Recency, Frequency and Monetary value (RFM), and used K-Means clustering to split them into groups that marketing can act on. After the dissertation, I added a Customer Lifetime Value (CLTV) model that turns those groups into a ranked list of actions.

What I found

- **Churn is the main problem.** About 45% of customers (the Lost and Attention Needed segments) had not bought anything for more than 1.5 years.
- **A tiny group carries a lot of value.** Four Champion customers (0.04%) were the biggest spenders in the whole base. With 13 clusters, the single top spender formed a cluster of their own.
- **Customers who buy more often spend more.** Frequency and Monetary were positively correlated in most segments ($r \approx 0.27 – 0.70$).
- **The business relies on one product line and two regions.** One flagship line held 47.6% of top-10 product units, and two regions made up about 76% of all transaction lines.
- **The data needed better upkeep.** 435 duplicate rows were hidden by product mis-tagging, 29% of records had no e-mail, and State names were typed inconsistently.

What I recommended

- Market to each segment differently: VIP care for Champion and Top Loyal customers, reactivation for Attention Needed, and low-cost win-back for Lost.
- Build visit frequency with refills, subscriptions and bundles, because frequency and spend move together.
- Cross-sell other product lines with the flagship, and expand into the 3rd to 5th largest regions.
- Fix product tagging and State values where the data is entered.
- With CLTV: protect high-value active customers, contact high-value customers who are drifting away first, and run a second-purchase programme for one-time buyers.

2 Introduction

Dissertation Chapter 1 · notebook: About this notebook

2.1 Background

Businesses now collect far more customer data than they can read by hand. Customer analysis turns that data into an understanding of what customers want, when they buy and why. Mohammed et al. (2020) group customer data analysis into five areas: churn, satisfaction, visits, purchases and segmentation. In their review of recent machine learning work, churn and segmentation were the two less popular research areas.

My industry partner is an established Malaysian retailer of health products. It sells through its own website and through branches, and its B2C transaction records were the basis of this project.

2.2 The problem

- Marketing managers found it hard to make informed decisions, because the database was large and complex.
- It was hard to see clear patterns in customer preferences and behaviour over time.
- Following up with customers was tedious, because there was no segmentation or filtering in place.
- Product data was messy: product labels were mixed up and had produced thousands of SKU variants, which made it hard to judge which product bundles worked.

2.3 Objectives and research questions

1. Explore and implement a machine learning model that segments customers with K-Means clustering.
2. Help the company understand its customers' behaviour, characteristics and preferences, for more effective marketing.
3. Find trends, patterns and other useful information in the data that support business growth.

Research questions: (1) How does a machine learning technique such as K-Means perform customer segmentation on the company's database? (2) How can that segmentation improve the understanding of customer behaviour, preferences and buying trends?

3 What earlier studies showed

Dissertation Chapter 2 · notebook: not applicable (background)

Machine learning methods fall into three broad groups. **Supervised learning** learns from labelled examples, for classification or regression. **Unsupervised learning** finds structure in unlabelled data, through clustering, association rules and dimensionality reduction. **Deep learning** uses multi-layer neural networks. Customer segmentation usually has no labels to learn from, which makes it an unsupervised, clustering problem.

Study	Method	Data	What I took from it
Hosseini & Shabani (2015)	RFM + K-Means across time intervals, DBI	500,000 B2B POS transactions, 7 months	Customer value changes over time; DBI is a practical way to choose k (11 clusters)
Kansal et al. (2018)	K-Means, Agglomerative, Mean-Shift	200 retail customers	K-Means scored best on Silhouette and Calinski-Harabasz
Nandapala & Jayasena (2020)	K-Means; Elbow, Silhouette, Gap statistic	Retail customers	A clear step-by-step process; k = 6 worked best
Punhani et al. (2021)	K-Means in RapidMiner, DBI	25,000 e-commerce rows	DBI chose k = 4; segmentation extended to products and payment methods
Park & Jun (2009)	Fast K-Medoids	Iris, Soybean and synthetic data	Medoids are less sensitive to outliers than means
Arora et al. (2016)	K-Means vs K-Medoids	10,000 transactions	K-Medoids is more robust; K-Means is simpler and faster
Brito et al. (2015)	K-Medoids, CN2-SD subgroup discovery	10,775 fashion e-commerce orders	Six product and six customer clusters; choosing k was time-consuming
Hsu et al. (2012)	Hierarchical clustering with a concept hierarchy of items	Library loan records	Combined Silhouette, C-index and Isolation index to choose clusters
Zakrzewska & Murlewski (2005)	K-Means vs DBSCAN vs two-phase	Bank customers	K-Means scales best; DBSCAN handles outliers but needs careful tuning
Salehinejad & Rahnamayan (2016)	Recurrent neural network on RFM	Retail customer sequences	About 80% success predicting next-step RFM values

Why K-Means. It is simple, fast, scales to tens of thousands of customers, and is the most common method in segmentation studies. Its known weaknesses are choosing k, sensitivity to outliers and the starting points. I handled the first with two validation indices (CHI and DBI), and looked at the other two with different scaling and repeated runs.

4 How I approached it

Dissertation Chapter 3 · notebook: About this notebook, §0

I followed CRISP-DM (Chapman et al., 2000) and adapted it into six steps:

1 Raw data collection › **2 Feature** engineering › **3 Feature selection** (RFM) › **4 Model** development › **5 Model** evaluation › **6 Data** visualisation

Step	What it involved
1. Raw data collection	B2C transaction export from the partner's website: 62,766 lines, 17 variables, saved as CSV/Excel
2. Feature engineering	Cleaning, handling noise and outliers, transforming fields. This was the most time-consuming part of the project.
3. Feature selection	Recency, Frequency and Monetary (RFM), a well-established way to describe customer value (Hughes, 1994; Wei et al., 2010; Chen et al., 2012)
4. Model development	K-Means: choose k, place centroids, assign each customer to the nearest centroid by Euclidean distance, recompute the centroids, and repeat until nothing changes
5. Model evaluation	Davies-Bouldin Index (lower is better) and Calinski-Harabasz Index (higher is better) to choose k
6. Data visualisation	Charts and segment profiles that explain the result to marketing

The two validation indices, in plain words

- **Davies-Bouldin Index (DBI)** (Davies & Bouldin, 1979) compares each cluster with the cluster most similar to it: how spread out the two are, relative to how far apart their centres are. Lower values mean tighter, better-separated clusters.
- **Calinski-Harabasz Index (CHI)** (Calinski & Harabasz, 1974) is the ratio of between-cluster spread to within-cluster spread, adjusted for the number of clusters and points. Higher is better.

Tools

I used Python 3.9.7 in Jupyter (server 6.4.5) with NumPy, pandas, Matplotlib, SciPy, seaborn and scikit-learn. The original notebook was lost, so I later rebuilt it from the code and outputs in my report. It now runs on current library versions, with small compatibility edits that don't change the logic.

5 The data and how I cleaned it

Dissertation §4.1 – 4.3 · notebook §0, §0.1, §4.2, §4.3

The partner shared B2C transactions from 2017 to 2022: **62,766 lines and 17 features**, one line per product per invoice.

Type	Features	Notes
Nominal (7)	idMember, Name, Contact, Email, City, Invoice, Item_Name	Item_Name had 245 distinct products and bundles
Categorical (5)	Gender, Language, State, Profile_Type, Order_Type	Profile_Type had 16 product-line codes; Order_Type is Product or Bundle
Numerical (5)	Joined_Date, DOB, Invoice_Date, Quantity, Amount	Quantity: mean 2.60, std 4.10, max 140

5.1 Missing values (§4.3.1)

Feature	Missing	Share	What I did
Email	18,464	29.4%	Left as is: marketing reaches customers by phone
City	2,886	4.6%	Used to repair State where possible
Contact	283	0.5%	Left as is
State	53	0.1%	Repaired from City where possible, otherwise 'Unknown'. State mattered because the company wanted it for branch planning.
All other 13 features	0	0%	Complete

5.2 Noisy and inconsistent values (§4.3.2)

- **Language** had four labels for three languages. Two of them meant the same language, so I merged them, which left three labels.
- **State** had trailing spaces, two names for the same place, upper/lower-case variants, and catch-all values such as 'Others' or a blank. I wrote two functions that check each State against its City and merge everything into **19 clean labels**: Malaysian states, a few foreign countries, and 'Unknown'.

5.3 Hidden duplicates (§4.3.3)

A standard duplicate check found **zero** duplicate rows. The company told me there were some, though: the same item in one invoice was sometimes tagged under two product lines, which created extra rows that didn't look identical.

1. I built a dictionary of every invoice and its row positions. The longest invoice had **7** lines.
2. Within each invoice, I built a duplicate mask for each of Item_Name, Order_Type, Amount and Quantity, and combined the four masks.
3. Rows flagged on all four were collected and dropped.

> **Result:** 435 hidden duplicate rows removed, leaving **62,331** clean transaction lines. A standard check would have missed every one of them.

6 RFM features and K-Means modelling

Dissertation §4.4 – 4.5 · notebook §4.4, §4.5

6.1 Building RFM (§4.4)

- **Recency:** days since the customer's last purchase, measured from the last date in the data (early October 2022).
- **Frequency:** the number of transaction lines per customer. *In hindsight I'd count distinct invoices instead; the notebook keeps my original definition so the results match.*
- **Monetary:** total spend, the sum of Amount × Quantity.

Feature (9,227 customers)	Mean	Std	Min	Q1	Median	Q3	Max
Recency (days)	730.8	632.5	0	136	492	1,313	1,858
Frequency (lines)	6.76	13.32	1	1	3	6	248
Monetary (masked)	–	–	–	–	–	–	–

6.2 Scaling (§4.5.1)

The three features are on very different scales (days, counts and money), so I standardised them: y = (x − mean) / standard deviation. Frequency and Monetary stayed strongly right-skewed after scaling, with a few customers far out in the tail. Recency had two peaks: one group of recent buyers and one of long-lapsed buyers.

6.3 Choosing the number of clusters (§4.5.3)

I ran K-Means for k = 2 to 14 and scored each result with CHI and DBI. Each sweep took about 3 seconds.

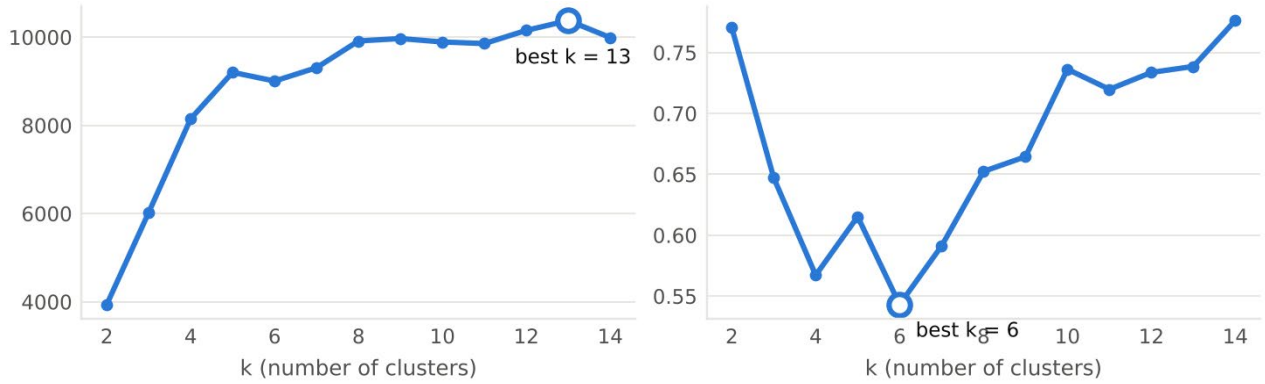


Figure 1. Standardised RFM. Left: Calinski-Harabasz Index (higher is better), highest at k = 13 (10,376.97). Right: Davies-Bouldin Index (lower is better), lowest at k = 6 (0.5431).

The two indices disagreed. With k = 13, the largest cluster held 2,656 customers (28.8%) and the smallest held just one, the top spender. **I chose k = 6.** Six segments are something a marketing team can actually act on, and more clusters don't by themselves make for better campaigns.

7 Who the customers are

Dissertation §5.1 · notebook §5.1

With k = 6, every customer falls into one of six groups. I profiled each group with summary statistics, distributions, pair plots, the Frequency–Monetary correlation and RFM quantile scores, then gave each one a business name.

The numbering from K-Means is arbitrary and changes between runs. The names below come from my dissertation run.

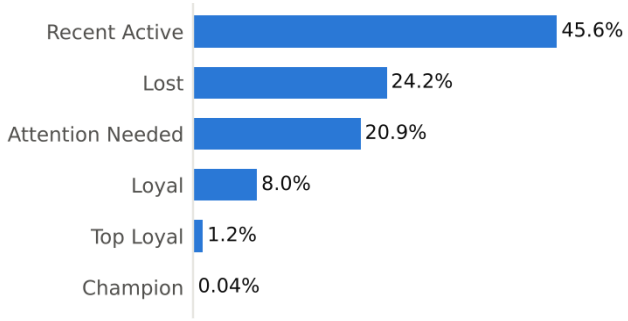


Figure 2. K-Means segments (k = 6), as a share of the 9,227 customers.

Segment	Customers	Share	What they look like	r (F, M)
Recent Active	4,204	45.6%	Bought within the last 619 days (75% within 329 days); 83% bought at least twice	0.70
Attention Needed	1,933	21.0%	Last purchase 613 – 1,399 days ago; most bought only a couple of times; mid spend	0.57
Lost	2,236	24.2%	Last purchase at least 1,321 days (3.6 years) ago; 70% bought more than once; low spend	0.65
Loyal	737	8.0%	75% bought within 342 days; every one at least 5 purchases; high spend	0.27
Top Loyal	113	1.2%	At least 24 purchases each; 75% bought within a year; very high spend	weak
Champion	4	0.04%	All bought within 175 days; 29 – 208 purchases; the highest spenders of all	n/a

What this means

- Almost half the base (Recent Active) is current but shallow. Moving these customers up to Loyal is the biggest growth lever.
- Lost and Attention Needed together are about 45% of customers. Win-back has to be cheap for Lost, and timely for Attention Needed, before they become Lost.
- Frequency and spend move together in most segments, so anything that brings customers back more often should raise revenue.
- Value is extremely concentrated: 4 customers sit far above everyone else and need personal, VIP treatment.

8 A second scaling: Standard + MinMax

Dissertation §5.2 · notebook §5.2

I wanted to see if the choice of k would hold up, so I added a MinMax normalisation on top of the standardisation and ran the same search again.

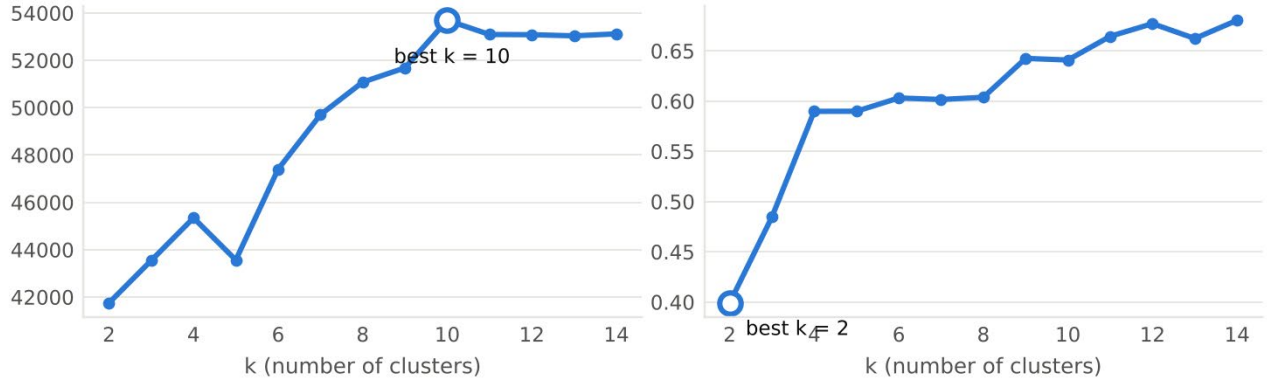


Figure 3. Standard + MinMax scaling. Left: Calinski-Harabasz Index, highest at k = 10 (53,691.91). Right: Davies-Bouldin Index, lowest at k = 2 (0.3991).

Choice	Cluster sizes (customers)
k = 10 (CHI)	1,011 · 2,225 · 846 · 906 · 835 · 99 · 551 · 394 · 1,037 · 1,323
k = 2 (DBI)	5,442 (59.0%) and 3,785 (41.0%)

- With this scaling, the best k changed much less between repeated runs.
- k = 10 gives a detailed view of customer types, while k = 2 gives a simple active-versus-lapsed split that is easy to act on.
- The cluster sizes are also more balanced than with standardisation alone, which is less dominated by outliers.

9 A rule-based alternative: RFM quantile scores

Dissertation §5.3 · notebook §5.3

Machine learning isn't always practical for a marketing team, so I also built a segmentation that runs on simple rules. Each customer gets a score from 1 to 4 for R, F and M, based on where they sit in the whole customer base. Score 4 is the best: most recent, most frequent, highest spend.

Score	Recency (days)	Frequency (lines)	Monetary
4 (best)	≤ 136	≥ 6	top quarter
3	137 – 492	3 – 5	second quarter
2	493 – 1,313	2	third quarter
1	> 1,313	1	bottom quarter

The three scores form a 3-digit code. The most common code was **444**, the best on all three, held by around 800 customers. Adding the scores gives an RFM total from 3 to 12. A small set of rules then maps each code to one of nine business categories:

Category	R	F	M
Champion	2 – 4	4	4
Top Loyal	3	any	3 – 4
Loyal	3	any	1 – 2
Top Recent	4	any	3 – 4
Recent Active	4	any	1 – 2
Top Customer At Risk	2 – 3	any	3 – 4
Attention Needed	2 – 3	any	1 – 2
Top Lost	1	any	3 – 4
Lost	1	any	1 – 2

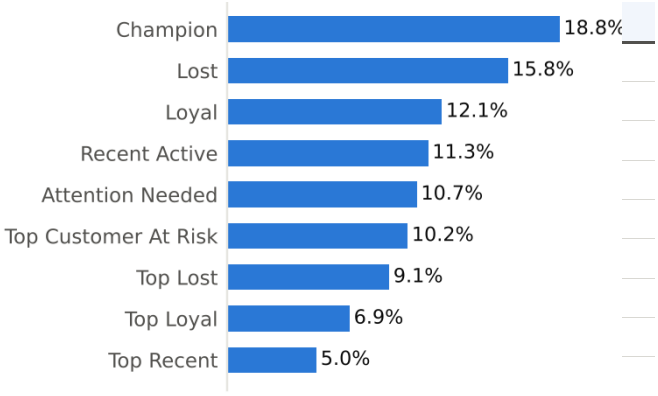


Figure 4. Rule-based categories, as a share of the 9,227 customers.

The groups are much more evenly sized than the K-Means clusters. The largest are Champion (18.75%) and Lost (15.84%), and the smallest is Top Recent (5.04%). Staff can apply the same rules to new data without running any model. *(The rules are applied in order, so a customer matching an earlier rule, such as Champion, is not counted again further down.)*

10 What they buy: product segmentation

Dissertation §5.4 · notebook §5.4

The catalogue has 16 product-line labels. Of the 62,331 clean lines, **53,882** were individual products and **8,449** were bundle lines, across **228** different bundle packages. I split individual-product sales into one column per product line and summed them by year.

10.1 Individual products

- One flagship line held **47.6%** of top-10 units. The top 3 lines held **73.5%**, and the other seven lines each had under 9%.
- The flagship's units jumped from 2017 to 2018–2019, dipped in 2020, and recovered through 2021–2022. *(2022 only runs to early October.)*

10.2 Bundles

- 228 different bundle packages were sold between 2017 and 2022. I ranked them to find the top 10 for marketing.
- In my dissertation I also wrote a 155-rule function to split each bundle back into its product lines. I didn't report its full output, so the notebook reproduces only the bundle rankings.

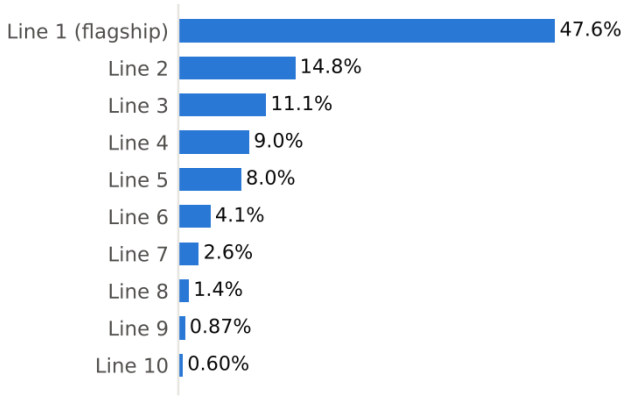


Figure 5. Top-10 product lines by share of units, 2017 – 2022. Names are masked.

What this means: the business depends heavily on one line. If that line slows, revenue slows with it. Bundling the flagship with the next lines is a low-risk way to spread demand.

11 Where they buy: regional analysis

Dissertation §5.5 · notebook §5.5

The company wanted to know where to open its next branch, so I compared regions by number of transaction lines and by sales. There were 18 regional labels, including a few foreign countries, plus 'Unknown' for lines whose location couldn't be determined.

- The HQ region (Region 1) and one branch region (Region 2) together made up **75.8%** of transaction lines and **74.9%** of sales.
- Region 1 accounts for a larger share of sales (48.1%) than of transaction lines (42.2%), so sales per line are higher there.
- The next three regions are much smaller, at about 3 – 4.5% of sales each.

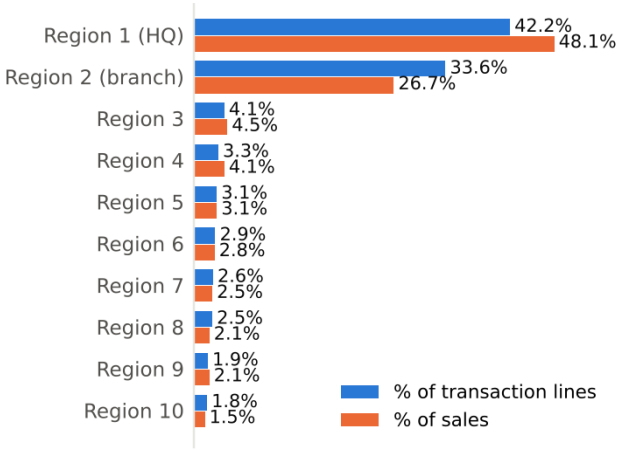


Figure 6. Top-10 regions, ranked by sales: share of transaction lines (blue) and of sales (orange). Region 1 is the HQ region and Region 2 a successful branch. Names are masked.

> **Recommendation:** the two largest regions are well covered. Expand next into Regions 3, 4 and 5, the 3rd to 5th largest by sales.

12 Conclusions of the dissertation

Dissertation Chapter 6 · notebook: Chapter 6

Contributions

- I segmented customers by how recently they visited (R), how often they bought (F) and how much they spent (M).
- I found that frequency and spending go together.
- I ranked customers from best to worst with a simple scoring categoriser.
- I segmented individual and bundle products.
- I supported an expansion plan with the regional analysis.

Limitations

- Clustering results change between runs, and there is no ground-truth label to check them against.
- Some customer inputs were inconsistent or doubtful, such as dates of birth that seemed entered just to fill the field.
- My business domain knowledge was limited, and so was the time available.

Future work suggested in the dissertation

- Collect more variables: login and logout times, referral links, discount values and basket contents.
- Try other clustering methods: K-Median, DBSCAN and hierarchical clustering.

After the dissertation I took one more step: estimating what each customer is worth in the future. That is the next chapter.

13 Further work: Customer Lifetime Value

Added after the dissertation · notebook §5.6

> **Status:** the method is complete and was tested on synthetic data with the same structure as the real data. Real-data CLTV results are pending access to the partner's data, so this chapter reports **no** real CLTV numbers.

13.1 What CLTV means

RFM tells me what a customer *has done*. **Customer Lifetime Value estimates what a customer is worth from now on**: the profit they are likely to bring over the rest of the relationship. A café regular who buys a coffee three times a week for years is worth far more than a tourist who spends more, but only once. CLTV puts one forward-looking number on each customer, so marketing can decide how much it's worth spending to keep them.

The simplest version (Gupta & Lehmann, 2003) is:

$$CLV = m \times r / (1 + i - r)$$

- *m*: the yearly margin from a customer
- *r*: the retention rate, the chance they're still a customer next year
- *i*: the discount rate, because money next year is worth a little less than money today

With 60% retention and a 10% discount rate, a customer is worth **1.2 years** of margin. At 70% retention that rises to **1.75 years**. Small gains in retention add up quickly. Reichheld & Sasser (1990) reported that, in the service businesses they studied, reducing defections by 5% raised profits by 25–85%.

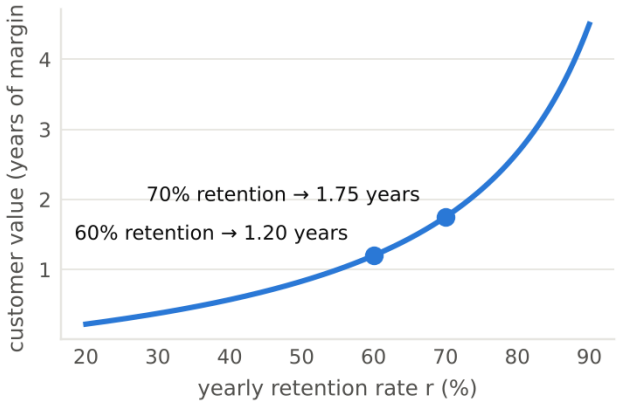


Figure 7. Simple CLV: customer value (in years of margin) by yearly retention rate, at a 10% discount rate. From the formula alone, with no company data.

13.2 The models

Retail customers don't cancel a contract. They just stop buying, and nobody announces it. Two standard models deal with this situation (Gupta et al., 2006):

Model	What it predicts	What it needs per customer
BG/NBD (Fader, Hardie & Lee, 2005a)	How many more purchases to expect, and the probability the customer is still active (P(alive))	Number of repeat purchases, time of the last purchase, time since the first purchase
Gamma-Gamma (Fader & Hardie, 2013)	Expected spend per purchase. Customers with few purchases are pulled towards the average, so one big order isn't over-trusted.	Number of repeat purchases and their average value

CLTV for the next 12 months = expected purchases × expected spend per purchase × profit margin, discounted month by month. I implemented both models directly from the published formulas with SciPy, with no black-box package. For CLTV, a purchase is one customer-day, which is different from the line-based Frequency in RFM, because these models need separate purchase occasions.

13.3 How I checked it

- **Hold-out test:** I fit the model on all but the last 26 weeks, then predicted those weeks. On the synthetic data, total predicted purchases were within about 2% of actual.
- **Gamma-Gamma assumption:** spend per purchase should not depend on how often someone buys. I check this with the correlation between purchase count and average spend, which was close to zero on the synthetic data.
- **Privacy:** CLTV runs on the camouflaged data and is shown only as an index (average customer = 100) and as shares, never in RM.

13.4 From numbers to actions

Group	Who	Action
Protect	High CLTV, probably still active (P(alive) ≥ 0.5)	VIP perks, early access, personal follow-up
Rescue now	High CLTV, probably drifting away (P(alive) < 0.5)	Win-back call or offer first, since there's the most value at stake
Grow	Lower CLTV, active	Bundles and cross-selling to raise frequency
Low-cost reactivation	Lower CLTV, drifting away	Automated reminders only
Second-purchase programme	One-time buyers	Follow-up offer or refill reminder after the usual repurchase gap

One-time buyers get their own group because of how BG/NBD works. It assumes customers can only drop out after a repeat purchase, so anyone who bought once always gets P(alive) = 1. I treat their CLTV as an upper bound.

13.5 Limitations

- The profit margin is an assumption (30% in the notebook). CLTV scales directly with it, so rankings and shares are reliable even if the absolute level isn't.
- BG/NBD assumes each customer buys at a steady rate until they stop, with no seasonality or campaigns.
- The camouflage shifts each customer's dates by up to ±30 days, so recency-based values are approximate. A private run can use a single global shift to keep them exact.

14 Business findings and recommendations

Notebook §5.7

14.1 Findings

#	Finding	Evidence
1	Six behaviour-based segments: Recent Active 45.6%, Lost 24.2%, Attention Needed 21.0%, Loyal 8.0%, Top Loyal 1.2%, Champion 0.04%	K-Means, k = 6 (§7)
2	Churn is the biggest issue: about 45% of customers hadn't bought anything for more than 1.5 years	Lost + Attention Needed (§7)
3	Value is extremely concentrated: the 4 Champions are the biggest spenders; with k = 13 the top spender formed a cluster of one	§6.3, §7
4	Customers who buy more often spend more (r ≈ 0.27 – 0.70 in most segments)	Per-segment correlation (§7)
5	A rule-based approach works too: nine balanced segments that staff can use without machine learning	Quantile scores (§9)
6	Standard + MinMax scaling gives more stable choices of k (10 and 2)	§8
7	Product dependency: one line held 47.6% of top-10 units, and the top 3 held 73.5%	§10
8	Geographic concentration: two regions made up 75.8% of lines and 74.9% of sales	§11
9	Data quality: 435 hidden duplicates, 29% missing e-mails, inconsistent State labels	§5

14.2 Recommended actions

#	Action	Expected effect
1	Market to each segment differently: VIP care for Champion and Top Loyal, reactivation for Attention Needed, low-cost win-back for Lost	Better response rates and a lower cost per sale
2	Set each customer's retention budget from their CLTV	Marketing money follows future value
3	Keep a 'Rescue now' list of high-value customers who are drifting away	Protects the largest share of future revenue
4	Run a second-purchase programme for one-time buyers	Turns a large group into repeat buyers
5	Offer refills, subscriptions and bundles that bring customers back more often	More visits mean more revenue
6	Bundle the flagship line with other lines	Less dependence on one line
7	Expand into Regions 3 to 5	Growth outside the two saturated regions
8	Fix product tagging and State values, and capture e-mail and consent, when data is entered	Cleaner data, so better models
9	Track retention, active share, total CLTV and the size of the Rescue-now list every month	Shows whether the actions work

15 How the data is protected

Notebook §0.1

The notebook is public, so the data is camouflaged straight after it is loaded, before anything is printed or plotted. The raw table is then deleted from memory.

Field	Camouflage	Why the analysis still works
Member ID	Integer pseudonym in a secret order (keyed HMAC-SHA256)	Joins, counts and RFM are unchanged
Name, phone, e-mail	Keyed tokens; missing values stay missing	Missing-value counts are unchanged
Invoice	Pseudonym such as INV-P000123	Invoice grouping and duplicate detection are unchanged
Dates	All of a customer's dates move by the same secret offset (± 30 days)	Gaps between purchases and tenure are unchanged
Date of birth	Reduced to a 5-year band	Not used in the analysis
City, State	Tokenised; states become Region A, B … after cleaning	Regional rankings and shares are kept
Products	Product lines coded P01 – P16; items ITEM-/BUNDLE-codes	Product and bundle analysis runs on the codes
Prices	Each product's price × its own secret factor (0.6 – 1.6)	Real prices can't be looked up
Business figures	Shown as percentage shares	Sales volumes stay private

- The pseudonyms can't be reversed without the secret key, which is kept offline and outside the project. Only the key holder can link a code back to a real customer, for example to act on a segment.
- Self-checks confirm that no real ID, invoice, name, phone, e-mail, product name, product code or price is left, and that row and customer counts are unchanged.
- In testing, clusters on the camouflaged data closely matched an uncamouflaged run (adjusted Rand index 0.965), which is within K-Means' normal run-to-run variation.
- Camouflage protects the customers, but the business still owns its data. Only what the NDA allows is published.

Appendix A: Report ↔ notebook map

Use this table to find the code behind each part of the report

Report section	Notebook section	What the code does
4 Approach	About this notebook · §0	Loads the real export, or creates a synthetic look-alike
15 Privacy	§0.1 Data camouflage	Pseudonymises customers, products, prices and dates; runs the self-checks
5 Data	§4.2 EDA	Feature types, date conversion, pair plot, summary statistics
5.1 Missing values	§4.3.1	Missing-value counts and State fill-in
5.2 Noise	§4.3.2	convertMajorLanguage, transformCITYInfo, transformSTATEInfo
5.3 Duplicates	§4.3.3	getDuplicatesWithInfo, detectPartialDuplicate, extractDuplicateIndex, getCleanData
6.1 RFM	§4.4	CustomerR, CustomerF, CustomerM merged into RFMdata
6.2 – 6.3 K-Means	§4.5	StandardScaler, CHI/DBI sweep for k = 2 – 14, k = 13 and k = 6 fits
7 Segments	§5.1	Per-cluster profiling and quantile Q-scores
8 Second scaling	§5.2	MinMax on top of standardisation; k = 10 and k = 2
9 Rule-based	§5.3	Recency/Frequency/Monetary_Scoring, RFM_Scoring, Categorizer
10 Products	§5.4	Product-line columns, yearly totals, top bundles
11 Regions	§5.5	Lines and sales by region
13 CLTV	§5.6	Simple CLV, BG/NBD and Gamma-Gamma, hold-out check, CLTV index, action matrix
14 Findings	§5.7	Dissertation findings, run-based findings, recommendations
12 Conclusions	Chapter 6	Summary of contributions, limitations and future work

Note: the notebook runs on synthetic data until the real export is supplied, so its numbers will differ from the dissertation figures in this report. Its 'Checking this run against my dissertation' table compares the two.

Appendix B: Key numbers fact sheet

All figures are from the dissertation's published results, masked for publication

Item	Value
Transaction lines (raw / clean)	62,766 / 62,331
Features	17 (7 nominal, 5 categorical, 5 numerical)
Period	2017 – early October 2022
Distinct products and bundles / product-line codes	245 / 16
Hidden duplicate rows removed	435 (longest invoice: 7 lines)
Missing values	Email 18,464 · City 2,886 · Contact 283 · State 53
Customers	9,227
Recency (days): median / max	492 / 1,858
Frequency (lines): median / max	3 / 248
Best k, standardised: CHI / DBI	13 (10,376.97) / 6 (0.5431)
Best k, standard + MinMax: CHI / DBI	10 (53,691.91) / 2 (0.3991)
k = 6 segment sizes	4,204 · 2,236 · 1,933 · 737 · 113 · 4
k = 13: largest / smallest cluster	2,656 / 1
k = 2 sizes	5,442 / 3,785
Rule-based: largest / smallest category	Champion 18.75% / Top Recent 5.04%
Frequency–Monetary correlation by segment	0.70 · 0.65 · 0.57 · 0.27 (one weak)
Individual-product lines / bundle lines / bundle packages	53,882 / 8,449 / 228
Flagship line / top-3 lines (share of top-10 units)	47.6% / 73.5%
Top-2 regions: share of lines / of sales	75.8% / 74.9%
Regional labels after cleaning	18 + 'Unknown'

Appendix C: References

Selected from my dissertation's reference list, plus the CLTV sources

Arora, P., Deepali, & Varshney, S. (2016). Analysis of K-Means and K-Medoids algorithm for big data. Procedia Computer Science, 78, 507–512.

Brito, P. Q., Soares, C., Almeida, S., Monte, A., & Byvoet, M. (2015). Customer segmentation in a large database of an online customized fashion business. Robotics and Computer-Integrated Manufacturing, 36, 93–100.

Calinski, T., & Harabasz, J. (1974). A dendrite method for cluster analysis. Communications in Statistics, 3(1), 1–27.

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). CRISP-DM 1.0: Step-by-step data mining guide.

Chen, D., Sain, S. L., & Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. Journal of Database Marketing & Customer Strategy Management, 19(3), 197–208.

Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. IEEE Transactions on Pattern Analysis and Machine Intelligence, 1(2), 224–227.

Fader, P. S., & Hardie, B. G. S. (2013). The Gamma-Gamma model of monetary value [Technical note]. http://www.brucehardie.com/notes/025/gamma_gamma.pdf

Fader, P. S., Hardie, B. G. S., & Lee, K. L. (2005a). "Counting your customers" the easy way: An alternative to the Pareto/NBD model. Marketing Science, 24(2), 275–284.

Fader, P. S., Hardie, B. G. S., & Lee, K. L. (2005b). RFM and CLV: Using iso-value curves for customer base analysis. Journal of Marketing Research, 42(4), 415–430.

Gupta, S., Hanssens, D., Hardie, B., Kahn, W., Kumar, V., Lin, N., Ravishanker, N., & Sriram, S. (2006). Modeling customer lifetime value. Journal of Service Research, 9(2), 139–155.

Gupta, S., & Lehmann, D. R. (2003). Customers as assets. Journal of Interactive Marketing, 17(1), 9–24.

Hosseini, M., & Shabani, M. (2015). New approach to customer segmentation based on changes in customer value. Journal of Marketing Analytics, 3(3), 110–121.

Hsu, F. M., Lu, L. P., & Lin, C. M. (2012). Segmenting customers by transaction data with concept hierarchy. Expert Systems with Applications, 39(6), 6221–6228.

Hughes, A. M. (1994). Strategic database marketing. Probus Publishing.

Kansal, T., Bahuguna, S., Singh, V., & Choudhury, T. (2018). Customer segmentation using K-means clustering. 2018 International Conference on Computational Techniques, Electronics and Mechanical Systems (CTEMS).

Kaufman, L., & Rousseeuw, P. J. (1986). Clustering large data sets. Pattern Recognition in Practice, 425–437.

Mohammed, M. M., Mohamed, N. A., Adam, A. A., Ahmed, S. S., & Saeed, F. A. (2020). Current directions and future research priorities of customer data analysis. Journal of Information Systems and Informatics, 2(2), 300–311.

Nandapala, E., & Jayasena, K. (2020). The practical approach in customers segmentation by using the K-Means algorithm. 2020 IEEE 15th International Conference on Industrial and Information Systems (ICIIS).

Park, H. S., & Jun, C. H. (2009). A simple and fast algorithm for K-medoids clustering. Expert Systems with Applications, 36(2), 3336–3341.

Punhani, R., Arora, V. P. S., Sabitha, S., & Kumar Shukla, V. (2021). Application of clustering algorithm for effective customer segmentation in e-commerce. 2021 International Conference on Computational Intelligence and Knowledge Economy (ICCIKE).

Reichheld, F. F., & Sasser, W. E. (1990). Zero defections: Quality comes to services. Harvard Business Review, 68(5), 105–111.

Salehinejad, H., & Rahnamayan, S. (2016). Customer shopping pattern prediction: A recurrent neural network approach. 2016 IEEE Symposium Series on Computational Intelligence (SSCI).

Wei, J.-T., Lin, S.-Y., & Wu, H.-H. (2010). A review of the application of RFM model. African Journal of Business Management, 4(19), 4199–4206.

Zakrzewska, D., & Murlewski, J. (2005). Clustering algorithms for bank customer segmentation. 5th International Conference on Intelligent Systems Design and Applications (ISDA'05).